Hand Gesture Description: Letter "ह"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held upright in a vertical position.
- **Fingers State:**
    - **Thumb:** The thumb is extended and positioned vertically alongside the outer edge of the index finger.
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are tightly folded down into the palm, forming a closed fist.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These should be aligned horizontally at the top of the palm.
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These should be positioned close to the base of the fingers or the middle of the palm.
    - **Landmark 4 (Thumb tip):** This should be the highest point of the gesture, pointing upward and level with or slightly above the folded index finger knuckle.

Hand Gesture Description: Letter "ᨠᨡ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Thumb:** The thumb is fully extended and pointed out to the side, away from the palm.
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are tightly folded down into the palm, forming a closed fist.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand, representing the furthest point from the palm on the thumb side.
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These should be clustered together and positioned close to the base of the fingers (MCP joints) or the center of the palm.
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These should form a horizontal line at the top of the palm area.

Feedback Tip for your App: The primary difference between "ᨠ" and **"ᨠᨡ"** is the position of the thumb. In "ᨠᨡ", the thumb must be extended outward. If the user keeps their thumb pressed against their index finger, your system should prompt: "Extend your thumb out to the side".

Hand Gesture Description: Letter "ฉ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Thumb:** The thumb is extended and pointed out to the side, away from the palm.
    - **Pinky Finger:** The pinky finger is fully extended and pointed straight upward.
    - **Index, Middle, and Ring Fingers:** These three fingers are tightly folded down into the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand.
    - **Landmark 20 (Pinky Tip):** This is the highest point of the gesture, pointing straight up.
    - **Landmarks 8, 12, and 16 (Fingertips):** These should be clustered together and positioned close to the center of the palm.

Feedback Tip for your App: The key feature of **"ฉ"** is the combination of the extended thumb and the extended pinky while the middle fingers remain folded. If the user fails to extend the pinky, the system should prompt: "Extend your pinky finger upward".

Hand Gesture Description: Letter "ቕ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Thumb:** The thumb is fully extended and pointed out to the side, away from the palm.
    - **Pinky Finger:** The pinky finger is fully extended and pointed straight upward.
    - **Ring Finger:** The ring finger is also extended and pointed straight upward, parallel to the pinky.
    - **Index and Middle Fingers:** These two fingers are tightly folded down into the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand.
    - **Landmarks 20 (Pinky Tip) and 16 (Ring Finger Tip):** These are the highest points of the gesture, pointing straight up.
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These should be clustered together and positioned close to the center of the palm.

Feedback Tip for your App: The key difference between "ቖ" and **"ቕ"** is that in "ቖ," both the pinky and the ring finger must be extended. If only the pinky is extended, your system should prompt the user: "Extend your ring finger as well as your pinky" to achieve a high accuracy score.

Hand Gesture Description: Letter "ฮ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index Finger:** The index finger is fully extended and pointed straight upward.
    - **Middle, Ring, and Pinky Fingers:** These three fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over the fingernails of the folded middle and ring fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This is the highest point of the gesture, pointing straight up.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the palm, overlapping the middle finger (Landmark 12).
    - **Landmarks 12, 16, and 20 (Fingertips):** These should be clustered together and tucked toward the center of the palm.

Feedback Tip for your App: The key feature of **"ฮ"** is the single extended index finger while the thumb is tucked across the palm. If the user extends the thumb or more than one finger, your system should prompt: "Only extend your index finger and keep your thumb folded across your palm" to maintain high accuracy.

Hand Gesture Description: Letter "ঠ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are fully extended and pointed straight upward, held together.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over the fingernails of the folded ring and pinky fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These are the highest points of the gesture, pointing straight up and positioned close to each other.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the palm, overlapping the ring finger (Landmark 16).
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together and tucked toward the center of the palm.

Feedback Tip for your App: The key difference between "ও" and **"ঠ"** is the number of extended fingers. While "ও" uses only the index finger, **"ঠ"** requires both the index and middle fingers to be extended together. If the user only extends one finger or keeps the two fingers apart, your system should prompt: "Extend both your index and middle fingers together".

Hand Gesture Description: Letter "ج"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, and Ring Fingers:** These three fingers are fully extended and pointed straight upward.
    - **Pinky Finger:** The pinky finger is tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over the fingernail of the folded pinky finger.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip), 12 (Middle Tip), and 16 (Ring Tip):** These are the highest points of the gesture, pointing straight up.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the base or center of the palm, overlapping the pinky finger (Landmark 20).
    - **Landmark 20 (Pinky Tip):** This should be tucked toward the center of the palm.

Feedback Tip for your App: The key feature of **"ج"** is having exactly three fingers (index, middle, and ring) extended while the pinky and thumb are folded. If the user extends the pinky or only two fingers, your system should prompt: "Extend your index, middle, and ring fingers together, and keep your pinky folded".

Hand Gesture Description: Letter "ج"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are fully extended and pointed straight upward, held together.
    - **Thumb:** The thumb is folded across the palm, resting over the fingernails of the folded fingers or tucked toward the center of the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These are the highest points of the gesture, pointing straight up and aligned horizontally.
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These should form a horizontal line at the top of the palm.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the palm area.

Feedback Tip for your App: The key feature of **"ج"** is the full extension of all four fingers while keeping them together. If the user spreads their fingers apart or fails to extend all four, your system should prompt: "Keep all four fingers extended and held together with your thumb folded".

Based on the video provided, here is the technical description of the hand gesture for the SLSL letter "ඵ" to support your MediaPipe-based feedback system:

Hand Gesture Description: Letter "ඵ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are fully extended and pointed straight upward.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is extended and pointed out to the side, away from the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These are the highest points of the gesture, pointing straight up.
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand, distinct from the fingers.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together and positioned close to the center of the palm.

Feedback Tip for your App: The key feature of "ඵ" is the combination of the extended thumb with the extended index and middle fingers. If the user keeps their thumb tucked or folds one of the top two fingers, your system should prompt: "Extend your thumb, index, and middle fingers while keeping the others folded".

Hand Gesture Description: Letter "ಲ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, and Ring Fingers:** These three fingers are fully extended and pointed straight upward.
    - **Pinky Finger:** The pinky finger is tightly folded down into the palm.
    - **Thumb:** The thumb is fully extended and pointed out to the side, away from the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip), 12 (Middle Tip), and 16 (Ring Tip):** These are the highest points of the gesture, pointing straight up.
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand, distinct from the fingers.
    - **Landmark 20 (Pinky Tip):** This should be tucked toward the center of the palm.

Feedback Tip for your App: The key feature of **"ಲ"** is the combination of the extended thumb with three extended fingers (index, middle, and ring). If the user fails to extend the thumb or the ring finger, your system should prompt: "Extend your thumb, index, middle, and ring fingers while keeping your pinky folded".

Hand Gesture Description: Letter "ঘ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are partially curved or curled inward toward the palm, but not closed into a tight fist.
    - **Thumb:** The thumb is extended upward and curved slightly toward the fingers, with the tip of the thumb moving toward the tip of the index finger to form a partial "O" or "C" shape.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip) and Landmark 8 (Index Tip):** These two points are positioned close to each other, creating the top curve of the gesture.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be curved downward and inward, roughly parallel to each other.
    - **Landmark 0 (Wrist):** Positioned at the base, forming the bottom of the curved hand shape.

Feedback Tip for your App: The key feature of **"ঘ"** is the rounded, open shape created by the thumb and curved fingers. If the user closes their hand into a tight fist or keeps their fingers completely straight, the system should prompt: "Curve your fingers and thumb to form a rounded shape" to increase the accuracy score.

Hand Gesture Description: Letter "𐌟"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** These four fingers are curved inward toward the center of the palm, but they do not touch the palm. They form a large, open "C" or "O" shape.
    - **Thumb:** The thumb is extended upward and curved toward the index finger, completing the rounded shape. The tip of the thumb is positioned near, but not necessarily touching, the tip of the index finger.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip) and Landmark 8 (Index Tip):** These points are the primary focus, positioned to create the upper curve of the circle.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These points should follow the curve of the index finger, staying roughly parallel to it.
    - **Landmark 0 (Wrist):** Forms the base of the circular gesture.

Feedback Tip for your App: The key feature of **"𐌟"** is the open, rounded space created between the thumb and the curved fingers. If the user makes a tight fist or leaves their fingers straight, your system should prompt: "Make a rounded 'O' shape by curving your fingers and thumb toward each other".

Hand Gesture Description: Letter "ක්"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are tightly folded down into the palm, forming a closed fist.
    - **Thumb:** The thumb is folded across the front of the fist, typically resting horizontally over the middle joints (PIP joints) of the index and middle fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These should form a horizontal line at the top of the visible hand area.
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These should be clustered together and positioned close to the center of the palm, obscured or semi-obscured by the thumb.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the fist, roughly between the index and middle finger landmarks (8 and 12).

Feedback Tip for your App: The letter **"ක්"** is represented by a standard closed fist. If the user leaves any fingers extended or if the thumb is pointing upward (like "ඊ"), the system should prompt: "Close your hand into a tight fist with the thumb folded across your fingers" to ensure a high accuracy score.

Hand Gesture Description: Letter "ඛ" (Khi)

- **Palm Orientation:** The palm is facing inward, directed toward the opposite side of the body rather than the camera.
- **Hand Position:** The hand is held vertically, with the side of the hand (pinky side) facing the camera.
- **Fingers State:**
    - **Index Finger:** Fully extended and pointing straight upward.
    - **Middle, Ring, and Pinky Fingers:** Tightly folded down into the palm.
    - **Thumb:** Extended upward and pressed against the side of the folded middle finger, positioned parallel to the extended index finger.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This is the highest point of the gesture.
    - **Landmark 4 (Thumb Tip):** Should be positioned near the base or middle joint (PIP) of the index finger.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be clustered near the center of the palm.

Feedback Tip for your App: The key to this gesture is the vertical index finger while the rest of the hand is closed. If the index finger is slanted or other fingers are partially open, the system should prompt: "Keep the index finger straight and other fingers tightly folded".

Hand Gesture Description: Letter "ᵭ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index and Middle Fingers:** Both fingers are fully extended and pointed straight upward, held together.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is extended and pointed out to the side, away from the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These are the highest points of the gesture, pointing straight up and aligned vertically.
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand, creating a distinct gap between the thumb and the palm.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together and positioned close to the center of the palm.

Hand Gesture Description: Letter "ಙ"

- **Palm Orientation:** The palm is facing inward toward the body (sideways orientation).
- **Hand Position:** The hand is held horizontally, with the index finger pointing across the front of the neck or upper chest area.
- **Fingers State:**
  - **Index Finger:** Fully extended and pointed horizontally across the body.
  - **Middle, Ring, and Pinky Fingers:** Tightly folded down into the palm, forming a partial fist.
  - **Thumb:** Folded down, pressing against the middle joint of the folded middle finger to secure the fist.
- **MediaPipe Landmark Key Points:**
  - **Landmark 8 (Index Tip):** This point should have a significantly higher horizontal (X) value relative to the wrist (if using the right hand) as it points toward the opposite side of the frame.
  - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be clustered together near the palm center.
  - **Landmark 4 (Thumb Tip):** Positioned close to the palm, overlapping the base of the index or middle finger.

Hand Gesture Description: Letter "ඊ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index Finger:** The index finger is fully extended but **curved** slightly at the top, forming a "hook" or partial "C" shape.
    - **Middle, Ring, and Pinky Fingers:** These three fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is extended and pointed out to the side, creating a clear gap between the thumb and the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This point should be lower than its usual "straight up" position because the finger is hooked/curved.
    - **Landmark 4 (Thumb Tip):** This should be extended horizontally or diagonally away from the hand.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be clustered together and positioned close to the center of the palm.

Hand Gesture Description: Letter "ಹ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index Finger:** The index finger is extended but **curved/bent** at the middle and top joints, forming a "hook" shape.
    - **Thumb:** The thumb is extended and pointed out to the side (horizontally), creating a wide gap between the thumb and the index finger.
    - **Middle, Ring, and Pinky Fingers:** These three fingers are tightly folded down into the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This point should be significantly lower than Landmark 6 (the finger's base joint) because the finger is hooked.
    - **Landmark 4 (Thumb Tip):** This should have a high horizontal (X) value difference from the palm center, showing it is fully extended outward.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be clustered together and positioned close to the center of the palm.

Hand Gesture Description: Letter "ಕ್"

- **Palm Orientation:** The palm is facing inward (toward the user's face/body) rather than the camera.
- **Hand Position:** The hand is held vertically in an upright position.
- **Fingers State:**
    - **Index Finger:** Fully extended and pointed straight upward.
    - **Middle, Ring, and Pinky Fingers:** All three fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over or tucked behind the folded fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This is the highest point of the gesture, pointing straight up.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These should be clustered together and positioned close to the center of the palm.
    - **Landmark 4 (Thumb Tip):** This should be tucked inward, showing a low horizontal distance from the palm center (Landmark 9).

Hand Gesture Description: Letter "ಪ"

- **Palm Orientation:** The palm is facing slightly inward (medially) or toward the opposite side, rather than directly at the camera.
- **Hand Position:** The hand is held at head level, appearing relatively compact.
- **Fingers State:**
    - **Index Finger:** The index finger is bent at the knuckles (MCP and PIP joints) to form a squared-off "hook" or "half-box" shape.
    - **Middle, Ring, and Pinky Fingers:** These three fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is tucked in, often resting against the side of the folded middle finger or the palm.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This point is significantly lower than its fully extended height and is positioned closer to the thumb, reflecting the bent/hooked posture.
    - **Landmarks 12, 16, and 20 (Middle, Ring, and Pinky Tips):** These are clustered tightly in the center of the palm.
    - **Landmark 4 (Thumb Tip):** Positioned close to the palm, showing minimal extension.

Hand Gesture Description: Letter "ඩ"

- **Palm Orientation:** The palm is facing slightly inward (medially) toward the center of the body, with the back of the hand visible to the side.
- **Hand Position:** The hand is held vertically at chest level in a relaxed but distinct posture.
- **Fingers State:**
    - **Index and Middle Fingers:** Both fingers are extended and pointing slightly upward and forward. They are held together, forming a "V" or "two-finger" shape.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over the middle joints of the folded ring and pinky fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These should be the highest points in the hand landmark set, positioned close together.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together near the center of the palm.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the hand, overlapping the ring finger (Landmark 16).

Hand Gesture Description: Letter "ಧ" (Dha)

- **Palm Orientation:** The palm is facing slightly inward (medially) toward the center of the body, so the side of the hand is most visible to the camera.
- **Hand Position:** The hand is held vertically at chest/shoulder level in a steady, upright position.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are extended upward and slightly forward. They are held together, forming a "double finger" column.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting on top of the folded ring and pinky fingers to hold them in place.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These should be the highest landmarks, positioned close together at a similar height.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered near the center-bottom of the palm.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the middle of the palm, overlapping the base of the ring finger (Landmark 13 or 14).

Hand Gesture Description: Letter "ಅ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position at shoulder level.
- **Fingers State:**
    - **All Fingers:** All four fingers (index, middle, ring, and pinky) are tightly folded down into the palm, forming a standard closed fist.
    - **Thumb:** The thumb is tucked **under** the folded index and middle fingers. Unlike a typical "punching" fist where the thumb is on the outside, for this letter, the thumb tip is hidden or pressed against the inside of the fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These are clustered tightly together and are not visible from the front as they are pressed into the palm.
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These form a relatively straight horizontal line at the top of the fist.
    - **Landmark 4 (Thumb Tip):** This should have a very low depth (Z-axis) value relative to the fingers, as it is tucked behind them.

Hand Gesture Description: Letter "ක්"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position at shoulder level.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are tightly folded down into the palm, forming a closed fist.
    - **Thumb:** The thumb is folded across the **front** of the other fingers. Specifically, it rests horizontally over the middle joints (PIP joints) of the index and middle fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 5, 9, 13, and 17 (MCP joints):** These should form a relatively straight horizontal line at the top of the visible hand.
    - **Landmark 4 (Thumb Tip):** This should be positioned near the center of the fist, roughly between the index and middle finger columns.
    - **Landmarks 8, 12, 16, and 20 (Fingertips):** These should be tucked away and positioned close to the center of the palm area.

Hand Gesture Description: Letter "ᦀ"

- **Palm Orientation:** The palm is facing inward (medially) toward the center of the body.
- **Hand Position:** The hand is held vertically, with the side of the hand (index finger side) facing the camera.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are extended together and pointed **horizontally** toward the side.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is fully extended and pointed **straight upward**, perpendicular to the index and middle fingers, forming an "L" shape with the hand.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip):** This is the highest point of the gesture, pointing straight up.
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These should have significantly higher horizontal (X) values (pointing outward) compared to the palm center.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together and positioned close to the center of the palm.

Hand Gesture Description: Letter "ඩ"

- **Palm Orientation:** The palm is facing inward (medially), toward the side of the head, so the side of the hand is facing the camera.
- **Hand Position:** The hand is held vertically near the side of the face.
- **Fingers State:**
    - **Index and Middle Fingers:** Both are fully extended. However, unlike "ඩ", they are pointed in different directions: the **index finger points straight up**, and the **middle finger points horizontally forward** (toward the camera's left/user's right).
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is fully extended and pointed **straight upward**, parallel to the index finger.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip) and Landmark 8 (Index Tip):** These should be the highest points, appearing close together and pointing vertically.
    - **Landmark 12 (Middle Tip):** This should have a significantly higher horizontal (X-axis) value compared to the wrist, as it points outward/forward.
    - **Landmarks 16 and 20 (Ring and Pinky Tips):** These should be clustered near the palm center.

Hand Gesture Description: Letter "ඞ"

- **Palm Orientation:** The palm is facing **inward** (medially), toward the side of the head, so the side of the hand (thumb side) is facing the camera.
- **Hand Position:** The hand is held vertically near the side of the face.
- **Fingers State:**
    - **Index and Middle Fingers:** Both are fully extended. The **index finger points straight up**, and the **middle finger points horizontally forward** (forming a right angle between them).
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is folded across the palm, resting over the folded ring and pinky fingers. It does **not** point up.
- **MediaPipe Landmark Key Points:**
    - **Landmark 8 (Index Tip):** This is the highest point of the gesture, pointing vertically.
    - **Landmark 12 (Middle Tip):** This should have a significantly higher horizontal (X) value (or depth value depending on angle) than the wrist, as it points forward.
    - **Landmarks 16 and 20 (Ring and Pinky Tips):** These should be clustered near the center of the palm.
    - **Landmark 4 (Thumb Tip):** This should be tucked in, showing low extension compared to the index finger.

Hand Gesture Description: Letter "ę"

- **Palm Orientation:** The palm is facing **inward** (medially) toward the center of the body, meaning the side of the hand (thumb side) is facing the camera.
- **Hand Position:** The hand is held vertically at chest or shoulder level.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are extended together and pointed **horizontally** toward the side.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded down into the palm.
    - **Thumb:** The thumb is fully extended and pointed **straight upward**, forming a right angle (90°) with the extended index/middle fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmark 4 (Thumb Tip):** This is the highest point of the gesture, pointing straight up.
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These points should have a high horizontal (X) value (pointing outward) while maintaining a similar vertical (Y) value to their respective knuckles.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered together and positioned close to the center of the palm.

Hand Gesture Description: Letter "ಚ"

- **Palm Orientation:** The palm is facing **downward** toward the ground, with the back of the hand facing the camera.
- **Hand Position:** The hand is held at chest/shoulder level, with the arm bent at a 90-degree angle so the hand is positioned horizontally in front of the body.
- **Fingers State:**
    - **Index and Middle Fingers:** These two fingers are fully extended together and pointed **downward** toward the floor.
    - **Ring and Pinky Fingers:** These two fingers are tightly folded into the palm.
    - **Thumb:** The thumb is folded across the palm, tucked behind or resting against the folded ring and pinky fingers.
- **MediaPipe Landmark Key Points:**
    - **Landmarks 8 (Index Tip) and 12 (Middle Tip):** These should be the **lowest** points of the hand landmarks (highest Y-values in the coordinate system), pointing vertically down.
    - **Landmarks 5 and 9 (MCP Joints):** These will be at the top of the hand structure.
    - **Landmarks 16 (Ring Tip) and 20 (Pinky Tip):** These should be clustered near the palm center.
    - **Landmark 4 (Thumb Tip):** Tucked inward toward the palm.

Hand Gesture Description: Letter "ฮ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** Held vertically at shoulder level.
- **Fingers State:**
    - **Index, Middle, and Ring Fingers:** These three fingers are fully extended upward and held close together (touching).
    - **Pinky Finger and Thumb:** The pinky finger is curled down, and the thumb is folded across the palm to touch or hold the pinky finger in place.

MediaPipe Landmark Key Points:

- **Landmarks 8, 12, and 16 (Index, Middle, Ring Tips):** These will be the highest points on the hand (minimum Y-coordinates). They should be roughly at the same height.
- **Landmark 20 (Pinky Tip) & Landmark 4 (Thumb Tip):** These two landmarks should have a very small Euclidean distance between them ($d \approx 0$), as they are touching in the center of the palm.
- **Horizontal Alignment:** The distance between Landmarks 8, 12, and 16 should be minimal, indicating the fingers are not spread apart.

Hand Gesture Description: Letter "ශ්"

Palm Orientation: The palm is facing forward, directly toward the camera.

Hand Position: The hand is held vertically in an upright position at shoulder level, slightly higher than the previous gesture.

Fingers State:

Index, Middle, Ring, and Pinky Fingers: All four fingers are curled down toward the palm, but unlike a tight fist, the tips of these fingers are brought down to meet the thumb.

Thumb: The thumb is tucked under the four curled fingers. In this specific gesture, the thumb tip (Landmark 4) is generally obscured or resting beneath the nails/middle joints of the fingers.

MediaPipe Landmark Key Points:

Landmarks 5, 9, 13, and 17 (MCP joints): These form the highest point of the hand, creating a curved "arch" rather than a perfectly flat line.

Landmark 4 (Thumb Tip): This should be positioned behind the fingers. In a 2D camera view, the Z-coordinate for Landmark 4 will be higher (further from the camera) than the finger landmarks (8, 12, 16, 20).

Landmarks 8, 12, 16, and 20 (Fingertips): These are curled inward. Their Euclidean distance to the wrist (Landmark 0) is significantly shorter than in an open palm.

Hand Gesture Description: Letter "ఙ"

Palm Orientation: The hand is rotated so the back of the hand (dorsum) is partially visible to the camera, or the hand is in a semi-profile view.

Hand Position: Held vertically at shoulder level, but the wrist is slightly angled inward.

Fingers State:

Index, Middle, Ring, and Pinky Fingers: All four fingers are curled into a tight, rounded fist.

Thumb: Unlike the previous letter where the thumb was tucked under or across, in this gesture for "ఙ", the thumb is pressed against the side of the index finger or tucked slightly deeper into the palm, making the "knuckle" side of the fist the primary focal point for the camera.

MediaPipe Landmark Key Points:

Landmarks 5, 9, 13, and 17 (MCP joints): Because the hand is angled, these landmarks will appear in a diagonal or vertical cluster rather than a horizontal line.

Landmark 4 (Thumb Tip): This landmark will be positioned laterally (to the side) of the index finger (Landmark 5) rather than resting on top of the middle joints.

Landmark 0 (Wrist): The relationship between the wrist and the MCP joints will show a slight lateral rotation (ulnar/radial deviation) compared to the straight-up position of "ఠ".

Hand Gesture Description: Letter "ಚ"

Palm Orientation: The palm is facing forward, directly toward the camera.

Hand Position: The hand is held vertically at shoulder level.

Fingers State:

Index, Middle, Ring, and Pinky Fingers: All four fingers are tightly curled into a solid fist.

Thumb: The thumb is wrapped across the front of the fingers. Unlike "ಠ" where the thumb rests horizontally across the middle, for "ಚ", the thumb is pulled more tightly across the index and middle finger nails/distal phalanges.

MediaPipe Landmark Key Points:

Landmarks 5, 9, 13, and 17 (MCP joints): These should form a flat, horizontal line at the top of the hand.

Landmark 4 (Thumb Tip): This landmark is positioned almost exactly in the center of the fist's width, specifically overlapping Landmark 10 (PIP joint of the middle finger).

Landmarks 8, 12, 16, and 20 (Fingertips): These are completely hidden behind the thumb or tucked into the palm.

Hand Gesture Description: Letter "ಹ"

Palm Orientation: The palm is facing inward (toward the user's body/left side), making the side of the hand visible to the camera.

Hand Position: The hand is held at shoulder level, but the fingers are pointed horizontally across the chest (pointing toward the opposite shoulder).

Fingers State:

Index and Middle Fingers: These two fingers are fully extended and held tightly together, pointing horizontally.

Ring and Pinky Fingers: These are curled tightly into the palm.

Thumb: The thumb is folded down, resting against the ring finger or tucked into the palm, keeping the profile of the hand flat.

MediaPipe Landmark Key Points:

Landmarks 8 (Index Tip) and 12 (Middle Tip): These will have the highest X-coordinate values (if using a right hand) or lowest (if left), as they are the furthest points from the wrist in a horizontal plane.

Y-Coordinate Alignment: Landmarks 5, 8, 9, and 12 should have very similar Y-coordinates, forming a parallel line to the floor.

Landmarks 16 (Ring Tip) and 20 (Pinky Tip): These will be clustered near Landmark 0 (Wrist) or Landmark 13 (Ring MCP), indicating they are retracted.

Distance Check: The Euclidean distance between Landmark 8 and Landmark 12 should be very small, confirming the fingers are closed together.

Hand Gesture Description: Letter "ঈ"

Palm Orientation: The palm is facing forward, directly toward the camera.

Hand Position: The hand is held vertically at shoulder level.

Fingers State:Index Finger: Fully extended, pointing straight upward.

Thumb: Fully extended, pointing horizontally outward (forming a right angle with the index finger).

Middle, Ring, and Pinky Fingers: Tightly curled down into the palm.

MediaPipe Landmark Key Points:

Landmark 8 (Index Tip): This will have the minimum Y-coordinate (the highest point on the screen).

Landmark 4 (Thumb Tip): This will have the maximum (for right hand) or minimum (for left hand) X-coordinate, positioned significantly far from the palm center.

Angle Check: The angle formed by landmarks [4, 2, 8] (Thumb Tip, Thumb CMC, Index Tip) should be approximately 90 degrees.

Landmarks 12, 16, and 20 (Other Tips): These should be clustered near the palm landmarks (5, 9, 13, 17), indicating they are folded.

Hand Gesture Description: Letter "ന്"

Palm Orientation: The palm is facing forward, directly toward the camera.

Hand Position: Held vertically at shoulder level.

Fingers State:

Index Finger and Thumb: The tips of the index finger and thumb are touching, forming a circle (similar to the "OK" sign).

Middle, Ring, and Pinky Fingers: These three fingers are extended upward and slightly spread apart.

MediaPipe Landmark Key Points:

Landmark 4 (Thumb Tip) & Landmark 8 (Index Tip): The Euclidean distance between these two points should be near zero ($d \approx 0$). This is the primary trigger for this letter.

Landmarks 12, 16, and 20 (Middle, Ring, Pinky Tips): These should have significantly lower Y-coordinates (appearing higher on screen) than the circle formed by the index and thumb.

Vertical Alignment: Landmark 12 (Middle finger tip) will typically be the highest point (minimum Y-value) of the gesture.

Hand Gesture Description: Letter "ঞ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Index Finger:** Fully extended, pointing straight upward (90° relative to the ground).
    - **Thumb:** Fully extended, pointing horizontally outward (0° or 180° relative to the ground).
    - **Middle, Ring, and Pinky Fingers:** Tightly curled down into the palm, held in place.

MediaPipe Landmark Key Points:

- **Landmark 8 (Index Tip):** This point should have the lowest **Y-coordinate** (top of the frame).
- **Landmark 4 (Thumb Tip):** This point should have a high horizontal displacement from the palm (X-coordinate).
- **L-Shape Angle:** The angle between the vector **(Landmark 5 to 8)** and **(Landmark 2 to 4)** should be approximately **90°**.
- **Finger Extension State:**
    - is_extended(8) = True
    - is_extended(12, 16, 20) = False

Hand Gesture Description: Letter "ঙ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Index and Middle Fingers:** Both fingers are extended and **crossed** over each other. Usually, the middle finger (Landmark 12) crosses behind or over the index finger (Landmark 8).
    - **Ring and Pinky Fingers:** Tightly curled into the palm.
    - **Thumb:** Folded over the curled ring and pinky fingers to keep them tucked.

MediaPipe Landmark Key Points:

- **Landmark 8 (Index Tip) & Landmark 12 (Middle Tip):** These two points will be very close together horizontally. In a "crossed" state, their typical X-coordinate order is swapped (e.g., if using the right hand, the middle finger tip may move to the left of the index finger tip).
- **Depth (Z-axis):** MediaPipe's 3D landmarks will show one finger having a slightly higher Z-value (further from camera) than the other at the point of intersection.
- **Landmarks 16 (Ring) & 20 (Pinky):** Must be identified as "folded" (distance to wrist is small).

Hand Gesture Description: Letter "ශ්"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Thumb:** Fully extended, pointing horizontally outward.
    - **Pinky Finger:** Fully extended, pointing vertically or slightly outward.
    - **Index, Middle, and Ring Fingers:** Tightly curled into the palm.

MediaPipe Landmark Key Points:

- **Landmark 4 (Thumb Tip):** This should have a high horizontal displacement from the palm center.
- **Landmark 20 (Pinky Tip):** This will be the highest point on the outer edge of the hand (minimum Y-coordinate for that side).
- **Landmarks 8, 12, and 16 (Index, Middle, Ring Tips):** These should be clustered near the center of the palm, showing as "folded."
- **Distance Check:** The Euclidean distance between Landmark 4 and Landmark 20 should be near the maximum possible for the user's hand, confirming the wide stretch.

Hand Gesture Description: Letter "ᘓ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Index, Middle, and Ring Fingers:** These three fingers are folded down into the palm.
    - **Thumb:** The thumb is tucked **underneath** the index, middle, and ring fingers. Specifically, the thumb tip is placed between the ring and pinky fingers (near the base).
    - **Pinky Finger:** Curled down over the side of the thumb or held tightly against the ring finger.

MediaPipe Landmark Key Points:

- **Landmark 4 (Thumb Tip):** This is the most important differentiator. It should be positioned between **Landmark 13 (Ring MCP)** and **Landmark 17 (Pinky MCP)**.
- **Landmarks 5, 9, 13 (MCP joints):** These will form a horizontal line, but the fingers (8, 12, 16) will be resting on top of the thumb.
- **Z-Coordinate (Depth):** Landmark 4 will have a higher Z-value (further from the camera) than the folded fingers (8, 12, 16), which are "on top" of it.

Hand Gesture Description: Letter "ಹ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position at shoulder level.
- **Fingers State:**
    - **All Four Fingers (Index, Middle, Ring, Pinky):** Fully extended and held tightly together (no gaps between fingers).
    - **Thumb:** Extended but tucked tightly against the side of the palm/index finger. It does not stick out to the side.

MediaPipe Landmark Key Points:

- **Landmarks 8, 12, 16, and 20 (Fingertips):** These will all have low **Y-coordinates**, indicating they are at the top of the hand.
- **Landmark 4 (Thumb Tip):** This should be positioned very close to **Landmark 5 (Index MCP)** or the side of the index finger. The horizontal distance ($|X_4 - X_5|$) should be minimal.
- **Finger Parallelism:** The vectors for all four fingers (e.g., from MCP to Tip) should be nearly parallel, pointing toward the top of the frame.
- **No Gaps:** The Euclidean distance between adjacent fingertips (8 to 12, 12 to 16, etc.) must be below a specific threshold to confirm the hand is "closed" rather than "open/spread."

Hand Gesture Description: Letter "ඛ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Index, Middle, Ring, and Pinky Fingers:** All four fingers are fully extended and held tightly together, pointing straight up.
    - **Thumb:** The thumb is folded **across the palm**. Unlike "ඟ" where the thumb is tucked against the side, for **"ඛ"**, the thumb moves inward so the tip (Landmark 4) rests near the center of the palm or the base of the middle finger.

MediaPipe Landmark Key Points:

- **Landmarks 8, 12, 16, and 20 (Fingertips):** High vertical position (minimum Y-coordinates). They should remain close together.
- **Landmark 4 (Thumb Tip):** This is the "Key Feature." Its X-coordinate will be significantly closer to the center of the palm (near Landmark 9 or 13) than it was in the "ඟ" gesture.
- **Thumb Flexion:** The distance between Landmark 4 and Landmark 0 (Wrist) will be shorter than in the previous gesture, indicating the thumb is "closed" across the palm.

Hand Gesture Description: Letter "ஷ"

- **Palm Orientation:** The palm is facing **downward** toward the ground, with the back of the hand facing the camera.
- **Hand Position:** The arm is raised so the wrist is at shoulder level, but the hand is flexed downward.
- **Fingers State:**
    - **Index Finger:** Fully extended horizontally, pointing toward the opposite side of the body.
    - **Middle, Ring, and Pinky Fingers:** These three fingers are folded down vertically at a 90° angle from the palm, pointing toward the floor.
    - **Thumb:** Tucked behind or against the palm, out of the primary line of sight of the camera.

MediaPipe Landmark Key Points:

- **Landmark 8 (Index Tip):** This will be the furthest point horizontally from the wrist (maximum or minimum X depending on the hand used).
- **Landmarks 12, 16, and 20 (Middle, Ring, Pinky Tips):** These will have the highest **Y-coordinates** (appearing lowest on the screen) as they point toward the ground.
- **The "Bent" Angle:** The angle at the knuckles (MCP joints 9, 13, and 17) will be approximately 90°, while the Index MCP (Landmark 5) remains relatively flat.
- **Orientation:** Unlike most letters where the fingertips are the highest point, here the **Wrist (Landmark 0)** and **Knuckles (Landmarks 5-17)** will be higher on the screen than the fingertips.

Hand Gesture Description: Letter "ಪ"

- **Palm Orientation:** The palm faces **downward** toward the ground, with the back of the hand (dorsum) facing the camera.
- **Hand Position:** The wrist is held at shoulder level. The hand is flexed at the knuckles so the fingers point toward the floor.
- **Fingers State:**
    - **Index, Middle, and Ring Fingers:** These three fingers are extended and held together, pointing **straight down** toward the floor.
    - **Pinky Finger:** Curled into the palm or held tightly against the ring finger (often tucked).
    - **Thumb:** Tucked into the palm or held against the side of the hand, generally hidden from the camera's view.

MediaPipe Landmark Key Points:

- **Landmarks 8, 12, and 16 (Index, Middle, Ring Tips):** These will have the **highest Y-coordinates** in the hand landmark set, as they are the points closest to the bottom of the frame.
- **Vertical Alignment:** The X-coordinates for landmarks 8, 12, and 16 should be relatively close to their respective MCP joints (5, 9, 13), indicating a downward vertical drop rather than a horizontal extension.
- **Landmark 0 (Wrist) and MCPs:** These will be the "highest" points on the screen (lowest Y-values).
- **Landmark 20 (Pinky Tip):** Should show a significantly lower Y-coordinate (higher on screen) than the other three fingers, indicating it is retracted/folded

Hand Gesture Description: Letter "ع"

- **Palm Orientation:** The palm faces **inward** toward the user's face (or side-on to the camera).
- **Hand Position:** The hand is held at shoulder level with a vertical wrist.
- **Fingers State:**
    - **Index Finger:** Fully extended, pointing straight **upward**.
    - **Middle Finger:** Fully extended, pointing **horizontally** toward the center of the body.
    - **Ring and Pinky Fingers:** Tightly curled into the palm.
    - **Thumb:** Folded down, typically resting against the ring finger to keep the fist compact.

MediaPipe Landmark Key Points:

To programmatically detect this, you can use the following landmark relationships:

- **Landmark 8 (Index Tip):** This must have the lowest **Y-coordinate** (highest point on the screen).
- **Landmark 12 (Middle Tip):** This will have the highest **X-coordinate** (assuming a right-hand gesture) as it points toward the opposite shoulder.
- **The "90° Check":** The angle between the vector from **MCP 5 to Tip 8** (Index) and **MCP 9 to Tip 12** (Middle) should be approximately **90°**.
- **Fold Check:** Landmarks 16 (Ring Tip) and 20 (Pinky Tip) must have a distance to the wrist (Landmark 0) that is less than 50% of the index finger's length.

Hand Gesture Description: Letter "ඤ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically in an upright position at shoulder level.
- **Fingers State:**
    - **Index and Middle Fingers:** Folded down over the thumb.
    - **Thumb:** The thumb is tucked **between the middle and ring fingers**.
    - **Ring and Pinky Fingers:** Folded down into the palm, behind or beside the thumb.

MediaPipe Landmark Key Points:

- **Landmark 4 (Thumb Tip):** This is the high-priority feature. It should be positioned between **Landmark 10 (Middle PIP)** and **Landmark 14 (Ring PIP)**.
- **Z-Coordinate (Depth):** Landmark 4 will be "sandwiched" between the fingers. Its Z-value will be higher (further from camera) than the knuckles of the index/middle fingers but lower than the palm itself.
- **Landmarks 5, 9, 13, and 17 (MCP joints):** These form a relatively straight horizontal line, similar to other fist-based letters.

Hand Gesture Description: Letter "ඌ"

- **Palm Orientation:** The palm is facing forward, directly toward the camera.
- **Hand Position:** The hand is held vertically at shoulder level.
- **Fingers State:**
    - **Index Finger:** Fully extended, pointing straight upward.
    - **Middle, Ring, and Pinky Fingers:** Tightly curled into the palm to form a base.
    - **Thumb:** Folded across the middle and ring fingers to hold the fist shape secure.

MediaPipe Landmark Key Points:

- **Landmark 8 (Index Tip):** This will be the highest point of the hand (minimum Y-coordinate).
- **Landmark 4 (Thumb Tip):** Positioned near the center of the fist, overlapping the middle finger's proximal phalanx (Landmark 10).
- **Verticality:** The vector from Landmark 5 (Index MCP) to Landmark 8 (Index Tip) should have a very small X-displacement, confirming it is pointing straight up.
- **Folded State:** The Euclidean distance between the tips of the other three fingers (12, 16, 20) and the wrist (Landmark 0) should be minimal.